Project Plan

Team: Lucas Alves Peralta, Neel, Rohan

Project name: **Embedded traffic-light controller**

Who does what

Person	Writes these sections	Extra job
Lucas	1. Project scope and deliverables 2. Team objectives	Takes notes at meetings and writes the meeting summary (section 6)
Neel	3. Technologies and tools	Sets up the GitHub repo and uploads the final PDF
Rohan	4. Project timeline 5. Communication plan	Puts the final PDF together and makes it look clean
Everyone	Own individual objectives, own row in the contribution table (section 7)	Own reflection and peer evaluation (submitted separately)

1. Project scope and deliverables

Owner: Lucas

What is the project, what will it do, and what will we hand in at the end?

Our project is an embedded traffic-light controller written in C++. The program simulates a traffic light at an intersection. It cycles through red, green, and yellow using set times for each light. It also has an input menu with a pedestrian button, so a user can request to cross and the light changes safely. The timing can be changed, and we will test edge cases like pressing the button during yellow or pressing it more than once.

2. Objectives

Owner: Lucas (team objectives). Everyone writes their own individual objectives.

Keep every objective clear, realistic, and measurable. Example: "Finish the input menu and test it with 5 cases by Week 3."

Team objectives

1. Have a working program that cycles red, green, and yellow correctly by Week 3.
2. Have the pedestrian button and input menu working and tested with at least 5 cases by Week 3.
3. Make the timing configurable, test the edge cases, and merge everything to main by Week 4.
4. Track every task as a GitHub Issue with an owner and due date, and close it when it's done.
5. Have the final PDF and code in the repo by the due date.

Person	My objectives	How I'll contribute
Lucas	Write the project scope and team objectives by end of Week 1. Build the input menu and pedestrian button and	Project scope and goals, input and pedestrian button code, helping with the core logic, meeting notes.

Person	My objectives	How I'll contribute
	test them with 5 cases by Week 3. Help Rohan build the core light cycle by Week 3. Take notes at every weekly meeting.	
Neel	Set up the GitHub repo with a README, folder structure, and all teammates added by end of Week 1.	Repo setup and Git help for the team
Rohan	Finish the project timeline and communication plan by end of Week 1. Implement and test the core light cycle by Week 3. Deliver the final formatted PDF by Week 5.	Planning and scheduling, core logic, final document

3. Technologies and tools

Owner: Neel

List what we'll use and one sentence on why it fits the project.

Tool / language / framework	What we'll use it for	Why we picked it
C++	Main program	It's what we're learning in class
GitHub	Storing code and the plan	Required, and it tracks everyone's work
VS Code	Writing and editing the code	It is free and it is what we are most used to using
GitHub Issues	Tracking tasks and bugs	Keeps the to do list in the same place as the code
g++ on Linux	Compiling and running the program	It matches the class environment

4. Project timeline

Owner: Rohan

Every task needs a person, a deadline, and a deliverable.

Task	Who	Deadline	Deliverable	Milestone?
Repo set up with README and folders	Neel	End of Wk 1	Repo link, all members added	Yes
Scope, objectives, tools sections	Lucas, Neel	End of Wk 1	Sections 1–3 drafted	
Design light states and timing (state diagram)	Rohan	Wk 2	Diagram in repo	
Core state machine (red/yellow/green cycle)	Rohan + Lucas	Wk 3	Compiles with g++, cycles correctly	Yes

Task	Who	Deadline	Deliverable	Milestone?
Input menu / pedestrian button handling	Lucas	Wk 3	Working input, tested with 5 cases	
Timing configuration and edge cases	Neel	Wk 4	Tested code merged to main	
Full testing and bug fixes	Everyone	Wk 4	Test results logged in Issues	Yes
Final PDF assembled and formatted	Rohan	Wk 5	One clean PDF	
PDF uploaded and links submitted	Neel, everyone	Due date	PDF in repo, links submitted	Yes

5. Communication plan

Owner: Rohan

Question	Our answer
How do we talk to each other? (group chat, Discord, etc.)	Group chat (iMessage or Discord) for quick updates. Reply within 24 hours
When do we meet?	Once a week, e.g. after class or at a set evening time, plus extra if needed before deadlines
How do we track progress? (GitHub issues, checklist, etc.)	GitHub Issues: each task is an issue with an owner and due date, closed when merged.
What do we do if someone is stuck or behind?	Post in the chat as soon as you're blocked. If a task is 2+ days late, we reassign or pair up at the next meeting

6. Meeting summary

Owner: Lucas

Date	Who was there	What we decided
Nov 6	Everyone	Repo set up, plan sections done, state diagram finished. Start the core light cycle.
Nov 13	Everyone	Red/yellow/green cycle working. Input menu and pedestrian button tested with 5 cases.
Nov 20	Everyone	Timing settings and edge cases done and merged to main. Start full testing.
Nov 27	Everyone	All testing and bug fixes done, results logged in Issues. Final PDF drafted.

7. Contribution table

Owner: Everyone fills in their own row

Person	What I completed this iteration	My responsibilities next iteration
Lucas	Wrote Section 1 (project scope and deliverables) and Section 2 (team objectives), helped choose the project idea and split up the work, and took notes and wrote the meeting summary.	Build and test the input menu and pedestrian button with 5 cases, work with Rohan on the core red/yellow/green cycle, and keep taking meeting notes.
Neel	Created the GitHub repo, wrote the README and .gitignore, set up the src/ and docs/ folders, added teammates as collaborators, and wrote Section 3 (Technologies and tools)	Write and test the timer module, review teammates' code, and help with Git issues
Rohan	Worked on planning the project. Specifically completed the project timeline and communication plan	Diagram in repo

Before we submit

☐ Export as one PDF and upload it to the GitHub repo (Neel)

☐ Everyone submits the GitHub link

☐ Everyone submits their own reflection: what I did, how I helped, one thing to improve

☐ Everyone does the peer evaluation